

TUNIR SIKDAR

Data Engineer

Kolkata, India | sikdartunir@gmail.com | +91-6290072095
linkedin.com/in/tunir-sikdar-679161121 | github.com/tunir7sikdar | tunir7sikdar.github.io

PROFESSIONAL SUMMARY

Data Engineer, with almost 6 years of experience architecting and scaling distributed data platforms on AWS and Azure. Skilled at translating ambiguous data problems, missing feeds, inconsistent vendor formats, historical gaps, into fault-tolerant, idempotent pipeline designs and Lakehouse architectures supporting high-volume, mission-critical workloads. Owns system design end-to-end, improving reliability, performance, and cost efficiency for Fortune 500 clients across insurance, retail, and e-commerce.

SKILLS

- **Languages:** Python, SQL, T-SQL
- **System Design:** Distributed systems, Lambda & Kappa architecture, Medallion architecture, SLA-driven orchestration, scalability & process optimization
- **AWS:** Lambda, Glue, S3, EC2, CloudWatch, Secrets Manager, SNS
- **Azure:** Databricks, Databricks Jobs and Workflows, Data Factory (ADF), Data Lake Storage (ADLS Gen2), Key Vault, Azure SQL Database, Event Hubs
- **Big Data:** Apache Spark, PySpark, Spark SQL, Apache Iceberg, Hadoop (basic)
- **Data Engineering:** ETL/ELT design, CDC logic, data modeling & normalization, entity relationship diagrams, Data Warehousing, Snowflake schema design, Data Vault 2.0, data governance, data lineage, data quality frameworks
- **DevOps & Tools:** Azure DevOps, Git, CI/CD
- **MLOps:** Basic exposure to model deployment pipelines and monitoring

EXPERIENCE

Data Engineer II | Deloitte USI - Kolkata, India

04/2025 - Present

- Led system design for an Insurance client on AWS, architecting ingestion, processing, and loading into Delta tables using CDC for optimal data handling at 4.2 TB of daily volume with zero data loss.
- Led and mentored a team of 5 engineers, aligning Finance and Ops stakeholders through conflicting priorities to overhaul commission payout processing from weekly to daily, cutting manual effort by ~90% and chargeback turnaround from a month to a day.
- Architected a fault-tolerant orchestration layer on AWS Lambda with automatic retry, SNS alerting, and full audit trails sustaining a 99%+ on-time SLA.
- Implemented PII data masking and column-level access controls using Databricks Unity Catalog for centralized governance, ensuring GDPR-compliant, secure RBAC-based data access across teams.

Firm Initiatives

- Built an event ingestion pipeline for a mobile app using Azure Event Hubs and Spark Structured Streaming, tracking user health metrics and usage history for stakeholders.

Engineer | Fractal Analytics - Pune, India

09/2023 - 03/2025

- Designed, tested, and delivered 24 end-to-end Big Data projects for a Retail client, utilizing Spark and Python on Azure for datasets exceeding 500 GB.
- Implemented data quality and validation checks across multiple ingestion layers, applying imputation logic to fix recurring data gaps in non-critical data and ensuring 100% compliance with data quality SLAs.
- Built automated monitoring frameworks for fault detection and risk mitigation, cutting manual error handling by 60% and re-architecting workflows onto Azure to save \$150K annually.

Application Developer | IBM India Pvt. Ltd. - Kolkata, India

02/2021 - 08/2023

- Developed metadata-driven ADF pipelines to ingest data from sources including SAP across 40+ tables, cutting development effort by 30% for a Manufacturing client.
- Modeled data into a snowflake schema using PySpark and built serving-layer views per business standards, powering Power BI reporting for stakeholders.
- Implemented error handling, log file maintenance, and alerting mechanisms, improving incident response time by 60%.

EDUCATION

B.Tech. in Electronics and Communication Engineering, Heritage Institute of Technology

08/2016 - 08/2020

GPA: 9.09/10.0

CERTIFICATIONS

- Databricks Certified Data Engineer Associate (2025)
- Fractal Certified PySpark Developer

AWARDS

- Outstanding Award & Applause Award, Deloitte USI | Star Award (Kaizen), Fractal Analytics | Manager Appreciation Award, IBM India Pvt. Ltd.